WORKED SOLUTION

STEP II 2011, Question 6

Pure • Integration

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s2-q6

The key idea

The integrand $[f'(x)]^2[f(x)]^n$ is one factor of f' short of being something we can integrate on sight, because $f'(x)[f(x)]^n$ is the derivative of $[f(x)]^{n+1}/(n+1)$. So we integrate by parts, splitting off that piece. The leftover integral contains f'', and the condition $f'' = kff'$ turns it back into the same easy shape, $f'f^m$. The applications then come down to choosing f and n so that the given integrand is $[f']^2 f^n$, and finding k.

Solution

The general result

Write f for $f(x)$, and so on. Integrate by parts with $u = f'$ and $\dfrac{\mathrm{d}v}{\mathrm{d}x} = f'f^n$, so that $\dfrac{\mathrm{d}u}{\mathrm{d}x} = f''$ and $v = \dfrac{f^{n+1}}{n+1}$ (fine, since $n \geq 1$ means $n+1 \neq 0$):

$$\begin{aligned} I = \int f' \cdot f'f^n \,\mathrm{d}x &= \frac{f'f^{n+1}}{n+1} - \frac{1}{n+1}\int f''f^{n+1}\,\mathrm{d}x \\ &= \frac{f'f^{n+1}}{n+1} - \frac{k}{n+1}\int f'f^{n+2}\,\mathrm{d}x \qquad \text{using } f'' = kff' \\ &= \frac{f'f^{n+1}}{n+1} - \frac{kf^{n+3}}{(n+1)(n+3)} + C. \end{aligned}$$

So

$$\boxed{I = \frac{f'(x)\,[f(x)]^{n+1}}{n+1} - \frac{k\,[f(x)]^{n+3}}{(n+1)(n+3)} + C}$$

which is an expression in terms of $f(x)$, $f'(x)$, k and n, as required.

Part (i)

For $f(x) = \tan x$ we have $f'(x) = \sec^2 x$ and

$$f''(x) = 2\sec x \cdot \sec x \tan x = 2\tan x \sec^2 x = 2f(x)f'(x),$$

so the condition holds with $k = 2$. The formula then gives

$$\int \sec^4 x \tan^n x \, \mathrm{d}x = \frac{\sec^2 x \tan^{n+1} x}{n+1} - \frac{2\tan^{n+3} x}{(n+1)(n+3)} + C. \tag{1}$$

Verification. Directly, using $\sec^2 x = 1 + \tan^2 x$ and the substitution $t = \tan x$, $\mathrm{d}t = \sec^2 x \, \mathrm{d}x$:

$$\int \sec^4 x \tan^n x \, \mathrm{d}x = \int (1+t^2) t^n \, \mathrm{d}t = \frac{t^{n+1}}{n+1} + \frac{t^{n+3}}{n+3} + C.$$

Rewriting the right-hand side of (1) with $\sec^2 x = 1 + t^2$ gives

$$\frac{(1+t^2)t^{n+1}}{n+1} - \frac{2t^{n+3}}{(n+1)(n+3)} = \frac{t^{n+1}}{n+1} + \frac{(n+3)-2}{(n+1)(n+3)} t^{n+3} = \frac{t^{n+1}}{n+1} + \frac{t^{n+3}}{n+3},$$

the same answer. So the result is verified for $f = \tan$.

The integral. Since

$$\frac{\sin^4 x}{\cos^8 x} = \frac{\sin^4 x}{\cos^4 x} \cdot \frac{1}{\cos^4 x} = \tan^4 x \sec^4 x = [f'(x)]^2 [f(x)]^4,$$

this is the case $n = 4$ of (1). Using the simplified form just found,

$$\int \frac{\sin^4 x}{\cos^8 x} \, \mathrm{d}x = \frac{\tan^5 x}{5} + \frac{\tan^7 x}{7} + C.$$

Part (ii)

? **Ask yourself** Which f should we take, and where does the $\sec^2 x$ come from?

The bracket $(\sec x + \tan x)$ is the obvious candidate for f, so try $f(x) = \sec x + \tan x$. Its derivative is

$$f'(x) = \sec x \tan x + \sec^2 x = \sec x (\tan x + \sec x) = \sec x \, f(x).$$

This is the key fact: differentiating f just multiplies it by $\sec x$. Hence

$$\begin{aligned} f''(x) &= \sec x \tan x \, f(x) + \sec x \, f'(x) = \sec x \tan x \, f + \sec^2 x \, f \\ &= \sec x \, f \, (\tan x + \sec x) = \sec x \, f^2, \end{aligned}$$

and since $f f' = f \cdot \sec x \, f = \sec x \, f^2$ too, we have $f'' = f f'$, so $k = 1$.

Also $[f'(x)]^2 = \sec^2 x \, [f(x)]^2$, so

$$\sec^2 x \, (\sec x + \tan x)^6 = \sec^2 x \, f^2 \cdot f^4 = [f'(x)]^2 [f(x)]^4,$$

which is the case $n = 4$, $k = 1$ of the general result:

$$\begin{aligned}\int \sec^2 x\,(\sec x + \tan x)^6\,\mathrm{d}x &= \frac{f'f^5}{5} - \frac{f^7}{35} + C = \frac{\sec x\, f^6}{5} - \frac{f^7}{35} + C \\ &= \frac{f^6}{35}(7\sec x - (\sec x + \tan x)) + C \\ &= \frac{(\sec x + \tan x)^6\,(6\sec x - \tan x)}{35} + C.\end{aligned}$$

Checks

- Differentiate the answer to (i): $\sec^2 x \tan^4 x + \sec^2 x \tan^6 x = \sec^2 x \tan^4 x(1 + \tan^2 x) = \tan^4 x \sec^4 x$.
- Numerically, Simpson's rule gives $\int_{-0.7}^{1.1} \sin^4 x / \cos^8 x\,\mathrm{d}x \approx 22.12954$, matching $\left[\frac{1}{5}\tan^5 x + \frac{1}{7}\tan^7 x\right]_{-0.7}^{1.1}$; and $\int_{-1}^{0.9} \sec^2 x(\sec x + \tan x)^6\,\mathrm{d}x \approx 133.68566$, matching the answer to (ii).
- The general formula agrees with quadrature for $f = \tan$ and every n from 1 to 6.

Common slips

- Choosing $u = f^n$ in the integration by parts: this produces $f^{n-1}[f']^3$, which is worse. Pair one f' with f^n so that it integrates exactly.
- Taking $k = 1$ for $\tan x$: $f'' = 2\sec^2 x \tan x$, so $k = 2$.
- In (ii), forgetting that $\sec^2 x$ must be absorbed as $[f']^2/f^2$, which is why the power drops from 6 to $n = 4$.

Beyond STEP

The condition $f'' = kff'$ is itself a differential equation: it integrates once to $f' = \frac{1}{2}kf^2 + c$, a first-order separable equation whose solutions include $\tan$ (with $k = 2$, $c = 1$) and $\sec x + \tan x$ (with $k = 1$, $c = \frac{1}{2}$). Recognising that an integrand becomes tractable because the function satisfies an ODE is a theme that recurs in first-year work on special functions and reduction formulae. Boyce and DiPrima, *Elementary Differential Equations*, covers separable and autonomous equations like this one.

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